Haris Khawja

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Education

University of Waterloo — Waterloo, Ontario
Bachelor of Software Engineering – GPA: 92% — *2024 – Present*

Technical Skills

Languages: C/C++, Python, SQL (PostgreSQL), Java, C#, Scala, TypeScript/JavaScript
Frameworks & Libraries: Spring Boot, Django, .NET, Node.js/Express.js, Flask, REST, Bazel, RabbitMQ, Django-RQ, Redis, Playwright, Jest, Mockito, React-Native, TanStack Query, Next.js, React.js
Data & Machine Learning: NumPy, Pandas, scikit-learn, Matplotlib
Cloud & Deployment: AWS, GCP, Vercel, Supabase, Docker
Developer Tools: CMake, Git, GitHub, GitLab, Postman, Figma, Expo, Snyk

Experience

Software Engineer — Toronto, Ontario
BMO Capital Markets, Data Cognition | *C#, .NET, SQL* — *Sep. 2026 – Present*
- Rebuilt BMO's internal expense procurement application for **2000+** regular users across Global Markets, adding locking to safely handle concurrent database writes
- Optimized SQL query plans to reduce API fetch time from **18s → 300ms**

Software Developer — Waterloo, Ontario
Arctic Wolf Networks | *Spring Boot, Bazel, RabbitMQ, TanStack Query, Jest, Mockito* — *Jan. 2026 – Apr. 2026*
- Refactored backend architecture to eliminate chatty microservice patterns, reducing latency from **2.7s → 730ms**
- Remediated **20+ high-severity vulnerabilities** identified by **Snyk**, including the Billion Laughs attack, ReDoS, and RCE, maintaining compliance ahead of recurring app security audits
- Built a custom test engine for in-memory PDF generation to eliminate transpilation conflicts; reworked the CI pipeline using **Bazel** to enable coverage reporting **(c8/Istanbul)**, increasing test coverage from **16%** to **80%**

Full-Stack & Machine Learning Engineer — Toronto, Ontario
Cache | *Django, Django-RQ, GCP, Playwright, React-Native, TypeScript* — *May 2025 – Aug. 2025*
- Secured endpoints and ensured TAC-compliance by implementing **SQIDs**, preventing internal ID exposure and protecting **1000+** confidential database records from attacks
- Automated clothing resale workflows by crosslisting items on resale marketplaces using asynchronous **Playwright** scripts for **CRUD** operations, managed via **Django-RQ**, which increased resale handling speed by **96%**
- Developed a **Django/Bootstrap** dashboard to manage Playwright automation flows, reducing staff clothing inventory processing time from **15 minutes to 7 seconds** per item
- Built AI-driven pipelines using **Google Gemini OCR** and an **OpenAI tool-enabled model** with dynamic web scraping to extract clothing attributes and calculate resale valuations from live marketplace data

Projects

QuectoMM | *C++20, Python, Pandas, Matplotlib, CMake* — Sep. 2026 – Present
- Engineered a **C++20** L3 order book engine replaying **2.1M LOBSTER** NASDAQ events across 5 tickers
- Designed a Genesis-volume approximation model for liquidity unobservable in top-10 data, quantifying unknown queue volume at the touch **(0–15%)** while keeping FIFO priority for observed orders
- Validated event-by-event against LOBSTER ground truth: $\geq$**99.94%** best bid/ask agreement and $\leq$**0.29%** depth error pre-reconciliation **(0.00% on INTC/MSFT)**, with **95–99.6%** of events requiring no correction
- Building a research framework to measure the marginal impact of microstructure signals on an **Avellaneda-Stoikov** market-making strategy

LetsCook | *Express.js, PostgreSQL, Supabase, Next.js, React.js, TypeScript* — Jan. 2025 – Jun. 2025
- Developed LetsCook, a full-stack web application where users improve cooking skills through community-driven recipe challenges, featuring a competitive points system to motivate continuous improvement
- Deployed on **Vercel** and **Google Cloud Run** with **30+** active users and over **200** site visits weekly
- Created request handling and authentication using **REST APIs**, **Express.js**, and **Supabase** with **RLS**